

Supplementary Material: Multicalibration for Unbiased Model-Based Prevalence Estimation

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SI Appendix

S1. Formal Definitions of Prevalence Estimation Methods

This section defines the prevalence estimation methods used in the simulation and the applications. The simulation compares seven of them (Figure S2); main-text Figure 1 shows four of them, with Classify & Count at two thresholds.

Setup. Let $h(X) \in [0, 1]$ denote the classifier’s probabilistic prediction for input X , with true label $Y \in \{0, 1\}$. The goal is to estimate the target prevalence $\pi^* = P^*(Y = 1)$ using only unlabeled target data $\{X_i^*\}_{i=1}^n$ and calibration parameters estimated from a labeled source dataset.

S1.1 Uncalibrated Averaging

$$\hat{\pi}_{\text{raw}} = \frac{1}{n} \sum_{i=1}^n h(X_i^*)$$

S1.2 Classify & Count

Given a threshold τ chosen on calibration data:

$$\hat{\pi}_{\text{CC}} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[h(X_i^*) \geq \tau]$$

In the simulation, we report Classify & Count at two thresholds: $\tau = 0.5$, the default decision rule, and the score quantile at which $\hat{\pi}_{\text{CC}}$ reproduces the prevalence of the calibration sample. The second is itself a correction fitted to the labeled sample. In the ACS application, τ is this

prevalence-matched threshold; in the Llama replication (Section S2), τ maximizes Youden’s J on the calibration set ($\tau \approx 0.80$). For Claude Opus, Classify & Count uses the Yes/No label directly.

S1.3 Rogan-Gladen (Adjusted Count)

$$\hat{\pi}_{\text{RG}} = \frac{\widehat{\pi}_{\text{CC}} - \widehat{\text{FPR}}}{\widehat{\text{TPR}} - \widehat{\text{FPR}}}$$

where $\widehat{\text{TPR}}$ and $\widehat{\text{FPR}}$ are estimated from calibration data at threshold τ (Rogan and Gladen 1978). In the simulation, $\tau = 0.5$; in the applications, τ is as for Classify & Count (Section S1.2).

S1.4 Probabilistic Adjusted Classify & Count (PACC)

$$\hat{\pi}_{\text{PACC}} = \frac{\bar{h} - \hat{\mu}_0}{\hat{\mu}_1 - \hat{\mu}_0}$$

where $\bar{h} = \frac{1}{n} \sum_i h(X_i^*)$, $\hat{\mu}_1 = \hat{\mathbb{E}}[h(X)|Y = 1]$, and $\hat{\mu}_0 = \hat{\mathbb{E}}[h(X)|Y = 0]$ are estimated from calibration data (González et al. 2017).

S1.5 SLD (EMQ)

The Saerens-Latinne-Decaestecker algorithm iterates:

1. Initialize $\hat{\pi}^{(0)}$ from the source prevalence.
2. E-step: Adjust posteriors for the new prior:

$$\tilde{h}_i^{(t)} = \frac{(\hat{\pi}^{(t)}/\pi_s) \cdot h(X_i^*)}{(\hat{\pi}^{(t)}/\pi_s) \cdot h(X_i^*) + ((1 - \hat{\pi}^{(t)})/(1 - \pi_s)) \cdot (1 - h(X_i^*))}$$

3. M-step: $\hat{\pi}^{(t+1)} = \frac{1}{n} \sum_i \tilde{h}_i^{(t)}$
4. Repeat until convergence (Saerens et al. 2002).

SLD requires h to be a calibrated posterior in the source population. We therefore run it on the isotonic-recalibrated scores $m(h(X))$ of S1.6, not on the raw scores, in the simulation and in all applications; on raw scores its bias would conflate miscalibration with the failure of the label-shift assumption.

S1.6 Global Calibration

Global calibration fits a monotone map m from scores to probabilities by isotonic regression of Y on $h(X)$ in the calibration data, and estimates $\hat{\pi}_{\text{iso}} = \frac{1}{n} \sum_i m(h(X_i^*))$. The same estimator is used in the simulation and in both applications. Because m depends on the score alone, it is calibrated at the calibration sample’s feature mix but not within feature-defined groups whose scores overlap.

S1.7 Multicalibration

In the simulation and in both empirical applications, we use MCGrad (Tax et al. 2026), a multicalibration algorithm based on gradient boosting. MCGrad operates in logit space: given a base predictor $f_0(X)$ with logit $F_0(X) = \text{logit}(f_0(X))$, it iteratively fits gradient boosted decision trees (GBDTs) on the residuals between labels and current predictions. At each round t , a GBDT g_t is trained with the current logit predictions as `init_score` and with the feature matrix consisting of the segment features (categorical and numerical) augmented by the current logit prediction as an additional input feature. The logit predictor is then updated as $F_{t+1}(X) = \alpha_t \cdot (F_t(X) + g_t(X))$, where α_t is a single scalar unshrinkage factor estimated by a one-parameter logistic regression of the labels on the combined logit $F_t(X) + g_t(X)$, applied to the full running logit rather than to the increment alone. Because α_t is fit to maximize fit of the combined logit to the labels at each round, it counteracts the shrinkage induced by the GBDT learning rate without disturbing the relative structure the trees discovered; Tax et al. (2026) give conditions under which the resulting sequence reduces multicalibration error. By including the prediction as a feature, GBDT splits naturally discover miscalibrated regions in the joint space of features and score levels, thereby approximating multicalibration without requiring explicit group specification. The number of rounds is chosen by cross-validated early stopping (five folds by default) to prevent overfitting. MCGrad uses LightGBM as the GBDT implementation. See Tax et al. (2026) for theoretical results and deployment details. Our claims are algorithm-agnostic: any procedure producing a multi-accurate, a fortiori multicalibrated, predictor over the feature class delivers the same guarantee, and several exist (Hébert-Johnson et al. 2018; Gopalan et al. 2022; Detommaso et al. 2024). An open-source implementation of MCGrad is available at <https://mcgrad.dev>.

S1.8 Multi-accuracy versus multicalibration

The main text states the universal-adaptability guarantee for a multicalibrated predictor and computes one with MCGrad. The property strictly required for prevalence estimation is weaker. Let $e(X, Y) = f(X) - Y$. A predictor is *multi-accurate* with respect to a group class $\mathcal{G}$ if

$$\mathbb{E}[\mathbf{1}\{X \in G\}e(X, Y)] = 0 \quad \text{for every } G \in \mathcal{G}.$$

This is equivalent to zero mean signed error within every group of positive probability. Multicalibration implies this condition by averaging $\mathbb{E}[Y \mid f(X) = v, X \in G] = v$ over the prediction values v within each group.

Let P and P^* denote the source and target populations, and let $r(X) = dP_X^*/dP_X$ be their density ratio. Under covariate shift and overlap,

$$\mathbb{E}^*[f(X) - Y] = \mathbb{E}[r(X)e(X, Y)].$$

Suppose the target shift is representable by the group class, meaning that for some $G_1, \dots, G_J \in \mathcal{G}$ and coefficients $a_1, \dots, a_J$,

$$r(X) = \sum_{j=1}^J a_j \mathbf{1}\{X \in G_j\}.$$

Multi-accuracy then gives

$$\mathbb{E}^*[f(X) - Y] = \sum_{j=1}^J a_j \mathbb{E}[\mathbf{1}\{X \in G_j\}e(X, Y)] = 0.$$

Thus one source-fitted predictor is unbiased for every target whose density ratio lies in the linear span of the calibrated group indicators (Kim et al. 2022). For a single fixed target, only the weaker condition $\mathbb{E}[r(X)e(X, Y)] = 0$ is necessary.

The same argument gives an approximate result. If a function g in the span of the group indicators approximates r , then

$$|\mathbb{E}^*[e(X, Y)]| \leq |\mathbb{E}[g(X)e(X, Y)]| + \mathbb{E}[|r(X) - g(X)|],$$

where the final term uses $|e(X, Y)| \leq 1$. The first term reflects remaining multi-accuracy error and the second reflects how well the calibrated group class represents the target shift.

Multicalibration remains the more useful practical target. It also controls calibration within score levels and therefore applies when selection or downstream analysis depends on the model’s own scores. MCGrad (Tax et al. 2026) approximates this property using the feature and score interactions learned by gradient-boosted trees (Section S1.7). Other procedures can provide the same population guarantee if they achieve the required group-level residual balance (Hébert-Johnson et al. 2018; Gopalan et al. 2022; Detommaso et al. 2024).

S1.9 Estimation and implementation details

Simulation. The data-generating process is given in the main text: $X \sim \text{Bernoulli}(1 - P(X = 0))$, $U \sim N(0, 1)$ independent of X , $P(Y = 1 \mid X, U) = \sigma(a_X + bU)$ with $a_0 = -2$, $a_1 = 1.5$, $b = 1.5$, and classifier score $h = \sigma(a_X + bU + \delta_X)$ with $\delta_0 = 0.8$, $\delta_1 = 0$. Each of the 50 runs draws a fresh calibration sample of $n = 10,000$ at $P(X = 0) = 0.5$, fits every estimator once, and evaluates bias and RMSE on 20 fresh unlabeled targets of $n = 10,000$ with $P(X = 0)$ evenly spaced in $[0.01, 0.99]$. Bias is relative to each target sample’s realized prevalence. MCGrad uses X as its single categorical feature with default hyperparameters. Definitions of all seven methods are in Section S1.

Comparative Agendas Project. The two campaigns (binary Yes/No and direct probability elicitation) were run separately to avoid anchoring. Benchmark implementations: Rogan-Gladen uses TPR/FPR estimated on the calibration set; IPW estimates the calibration-vs-target propensity with a default LightGBM classifier, the learner underlying MCGrad, on the same features MCGrad uses (country, document type, party, decade, and length), cross-fitted over five folds; isotonic regression is fit on the scores. Per-scenario prevalence error is in Table S2; the ReadMe comparison is in Section S7 and the Llama 3.3 70B replication in Section S2.

Claude Opus 4.6 elicitation. The CAP sample contains 30,000 documents, 5,000 from each of the six sub-populations. We classified them with Claude Opus 4.6 (Anthropic, model identifier `claude-opus-4-6`) on 2–3 April 2026, accessed through the Claude Code command-line interface rather than the API. A Claude Code session running Opus 4.6 split the sample into 300 shards of 100 documents and dispatched each shard to a sub-agent, which inherits the parent session’s model; 50 sub-agents ran concurrently, in six rounds per campaign. Each sub-agent classified the documents of its shard within a single context, so classifications within a shard are not strictly independent. Sampling parameters were Claude Code defaults and were not set by us. Texts were truncated to their first 4,000 characters. The two campaigns used separate sub-agents with the following prompts, where `{language}` is the document’s language, `{text}` the document, and `{codebook}` the CAP master-codebook definition of Law & Crime (Jones et al. 2025) given below.

Binary campaign: The following text is in `{language}`. Does it primarily fall under the policy topic “Law and Crime” as defined by the Comparative Agendas Project? `{codebook}` Text: `{text}` Answer Yes or No.

Probability campaign: The following text is in `{language}`. Consider whether it primarily falls under the policy topic “Law and Crime” as defined by the Comparative Agendas Project. `{codebook}` Text: `{text}` Without first deciding Yes or No, directly estimate the probability that this text is about Law and Crime. Provide $P(\text{Yes})$ and $P(\text{No})$ as your true belief probabilities, summing to 1.0. Respond in exactly this format: $P(\text{Yes}): [0.XX]$ $P(\text{No}): [0.XX]$

Codebook definition: Law and Crime includes: general law, crime, and family issues; law enforcement agencies including border, customs, and specialized enforcement agencies; white collar crime, organized crime, counterfeiting, fraud, cyber-crime, and money laundering; illegal drug crime and enforcement, criminal penalties for drug crimes, and international efforts to combat drug trafficking; court administration, bail, pre-release, fines, and legal representation; prisons, jails, and parole systems; juvenile crime and justice, and efforts to reduce juvenile crime and recidivism; child abuse, child pornography, sexual exploitation of children, and parental kidnapping; family issues, domestic violence, child welfare, and family law; domestic criminal and civil codes; crime control, prevention, and impact of crime; and police and domestic security responses to terrorism.

The score is the elicited $P(\text{Yes})$. One shard of 100 Belgian newspaper articles returned no binary answers and is excluded, leaving the 29,900 documents analyzed in the main

text. Because the model is proprietary and was run without a fixed seed, the classifications cannot be regenerated exactly; we release all model outputs, the prompt templates and the shard-management script, and the Llama 3.3 70B replication (Section S2) can be reproduced in full.

CAP data sources. All CAP files were downloaded from comparativeagendas.net on 2 March 2026: Danish parliamentary questions (Green-Pedersen and Mortensen 2019), Spanish oral questions (release 19.1) and El País and El Mundo front pages (Chaqués-Bonafont et al. 2019, n.d.), U.S. congressional bills (release 19.3) (Wilkerson et al. 2025), and Belgian television news and *De Standaard* newspaper stories (Walgrave et al., n.d.). None carries a DOI. As requested by the data providers: the data in the Danish Policy Agenda Project were collected by Christoffer Green-Pedersen and Peter Bjerre Mortensen with support from the Danish Social Science Research Council and the Research Foundation at Aarhus University. The Spanish data were originally collected by Laura Chaqués-Bonafont, Anna M. Palau and Luz M. Muñoz, with the collaboration of graduate students and the financial support of the Spanish Ministry of Innovation and Science and AGAUR. The Belgian data were collected by Stefaan Walgrave and his collaborators (Jeroen Joly, Anne Hardy, Brandon Zicha, Julie Sevenans, and Tobias Van Assche), with funding from the European Science Foundation (07-ECRP-008), the Flemish National Science Foundation (G.0117.11N) and the Belgian Federal Science Policy (IUAP P7/46). Neither the funders nor the original collectors of the data bear any responsibility for the analysis reported here.

American Community Survey. Training states are TX, MI, PA, OH, IL, GA, NC, VA (2016–2018); held-out test states are CA, NY, FL, WA, AZ, CO, with in-distribution test $n \approx 920,000$. Age-shifted targets are produced by importance-weighted resampling. Post-hoc calibration uses isotonic regression and MCGrad with categorical and numerical features. The 2016–2018 1-Year person files (U.S. Census Bureau 2019) were downloaded through folktabs (Ding et al. 2021) on 14 April 2026.

Uncertainty. Intervals for the CAP and ACS applications come from a nonparametric bootstrap that holds the classifier fixed (the Opus outputs; the ACS logistic regression), so they reflect sampling variability of the labeled calibration data and of the targets, conditional on the measurement device. Each replicate resamples with replacement the calibration set (for CAP, within each sub-population), the in-distribution test pool, and the out-of-distribution targets; redraws the reweighted scenarios from the resampled pools with a new seed; refits every estimator, including MCGrad, isotonic regression, the Rogan-Gladen rates and the IPW propensity models; and records the signed error against that replicate’s realized target prevalence. Calibration observations drawn more than once enter as a single row weighted by their count rather than as repeated rows: MCGrad selects its number of boosting rounds by cross-validation and IPW cross-fits its propensity model, and repeated rows would fall on both sides of a fold, favoring overfitted fits. Point estimates are from the original samples. We report 95% percentile intervals from 200 replicates for CAP (`cap_analysis/opus/bootstrap_cap_opus.py`) and 50 for ACS (`acs_analysis/bootstrap_acs.py`), where each replicate refits IPW on 20,000-observation targets against the 644,000-observation calibration set. Percentile intervals, rather than normal intervals, are needed because the IPW errors are heavy-tailed on the age-skewed ACS targets.

S2. Robustness: Replication with Open-Weight LLM (Llama 3.3 70B)

The main text reports results using Claude Opus 4.6 as the LLM measurement device. To verify that the findings are not specific to a particular model, we replicate the CAP analysis using Llama 3.3 70B Instruct (Meta 2024), a model in the Llama 3 family (Grattafiori et al. 2024) (4-bit NF4 quantized, run on a single A100 80GB GPU). We report results for verbalized confidence scores; we also describe token log-probabilities, which we considered and rejected.

S2.1 Score Extraction Methods

Log-probabilities. For each document, the model is prompted with the CAP codebook definition of Law & Crime and asked to respond Yes or No. The score is extracted from next-token log-probabilities: $h(X) = P(\text{Yes}) / (P(\text{Yes}) + P(\text{No}))$. This produces highly bimodal scores: 23% of the 105,000 documents score at exactly 0.0 or 1.0, and only 6% fall in the mid-range [0.1, 0.9]. Such scores carry little information about uncertainty, so we do not use them.

Verbalized confidence (2-stage). A two-stage dialogue first asks the model to classify the document (Yes/No), then asks it to estimate the probability that its answer is correct, following the verbalized-confidence approach of Tian et al. (2023), with an instruction we added to discourage degenerate outputs (“Note: a probability of exactly 0.0 or 1.0 would mean absolute certainty, which is rarely warranted”). The score is $P(\text{correct})$ if the answer is Yes and $1 - P(\text{correct})$ if No. This produces scores with negligible boundary mass (75% in [0.1, 0.9]) but only 11 unique score values.

S2.2 Data and Calibration

The Llama analysis uses the full 105,000-document sample (15,000 per sub-population for Denmark questions, Spain questions, U.S. bills, and Belgium newspaper; 30,000 for Spanish media; 15,000 for Belgian TV). The calibration set ($n \approx 40,000$) is drawn equally from the four in-distribution sub-populations. MCGrad is calibrated with categorical features (country, document type, party) and one numerical feature (decade). Because this analysis draws on the larger 105,000-document sample rather than the 29,900-document Opus main-text sample, the expert-coded true prevalences below differ slightly from Table S2 (e.g., 8.1% vs. 7.9% at baseline); these are sampling differences in the gold standard, not discrepancies in the method.

S2.3 Results: Verbalized Confidence Scores

Table S3 shows prevalence estimation error using Llama 3.3 70B with verbalized confidence scores. Unlike the main-text Opus headline, which multicalibrates the Yes/No label, MCGrad here multicalibrates the verbalized score.

Scenario	Shift Type	True Prev.	CC	RG	IPW	Iso.	MCGrad
Baseline	None	8.1%	+14.7	+0.6	+0.1	+0.2	+0.2
Country shift	Within-cal.	8.7%	+15.6	+2.2	-0.2	+1.3	+0.1
Doc-type shift	Within-cal.	6.5%	+16.0	+1.7	+0.0	+0.7	+0.1
Spain media	OOD doc type	19.3%	+15.4	+6.6	-11.1	-7.2	-4.9
Belgium TV	OOD doc type	11.1%	+13.3	-0.0	-2.6	-1.6	-3.4

Table S3: Prevalence estimation error (pp) for Law & Crime topic using Llama 3.3 70B with verbalized confidence scores. CC = Classify & Count, RG = Rogan-Gladen, IPW = importance-weighted estimation (cross-fitted gradient-boosted propensity model on country, document type, party, and decade), Iso. = isotonic regression.

The pattern is consistent with the main text’s Claude Opus results: MCGrad achieves near-zero error within the calibration distribution (≤ 0.2 pp) and degrades on OOD populations (-3.4 to -4.9pp). Several differences are notable:

- **Higher raw CC error** (+13 to +16pp vs. +2 to +5pp with Opus).
- **Comparable MCGrad within-calibration performance** (≤ 0.2 pp for Llama, ≤ 0.4 pp for Opus), confirming that multicalibration corrects for model-specific calibration errors.
- **Larger OOD error** on Spanish media (-4.9pp vs. Opus’s -1.9pp on binary labels / -4.5pp on probability scores), reflecting the combination of a weaker base model with a coarser score distribution. Note that this comparison mixes elicitation modes: the Llama numbers here use verbalized confidence (11 unique values), whereas the Opus main-text headline uses binary labels; the closest like-for-like comparison is Llama verbalized against Opus’s probability-score condition (43 unique values, SI Figure S3), and on that comparison the gap is consistent with the finer Opus score distribution. The broader point (that MCGrad’s within-calibration performance is near-identical across models and elicitation modes while the input score distribution matters chiefly out of support) holds in every comparison.

S2.4 Additional Baseline: SLD on Llama Scores

The SLD (EMQ) algorithm, designed for label shift rather than covariate shift, is run on the isotonic-recalibrated Llama scores (S1.5). It is accurate at baseline (+0.6pp) but errs under the within-calibration shifts (+5.4pp for the country shift, -1.4pp for the doc-type shift) and out of support (+3.0pp on Spanish media, +1.4pp on Belgian TV).

S3. Empirical results: ACS employment benchmark and detailed tables

As a check with exact ground truth, we estimate employment prevalence from American Community Survey microdata via the *folktables* package. The true rate in any subpopulation is known, and the classifier is an ordinary logistic regression rather than an LLM, so this confirms the correction is not specific to language models or to noisy gold labels. We predict employment from 16 sociodemographic features, training on eight states (2016–2018; approximately 1.5M observations) and calibrating on a held-out set ($n \approx 644,000$). Because employment rates vary sharply by age (76% for ages 25–54 versus 17% for 65+), we construct covariate shifts by resampling the test set to be young-skewed, old-skewed, or bimodal, yielding true employment rates from 12.8% to 46.0%; all calibration parameters are fixed across scenarios. We evaluate both on in-distribution states and on six held-out states.

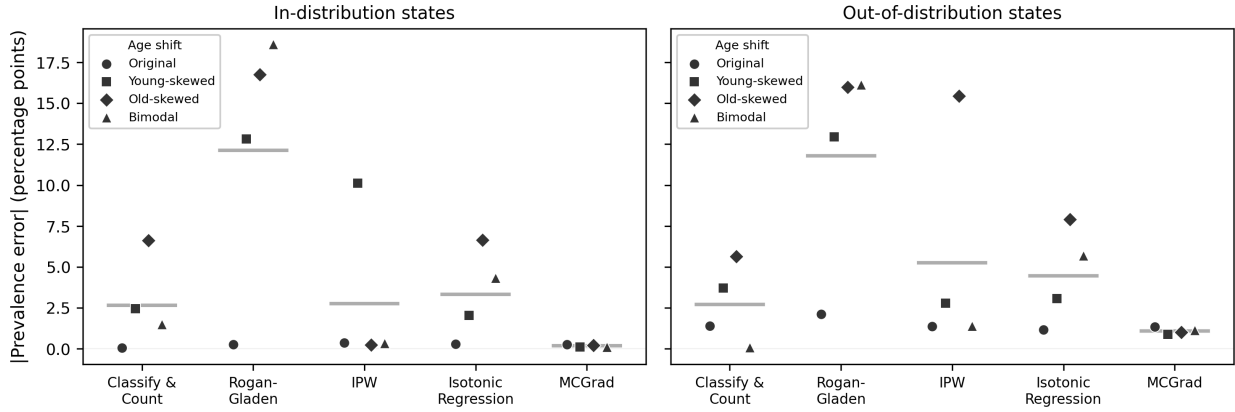


Figure S4. Absolute prevalence error (percentage points) for the ACS employment benchmark, by method and age-shift scenario, for in-distribution (left) and out-of-distribution (right) states. Marker shape denotes the synthetic age distribution; horizontal lines are per-method means. MCGrad is near-exact across all in-distribution scenarios and degrades only modestly out of distribution; Rogan-Gladen and isotonic regression grow with the age shift, and IPW collapses on two targets where its weights degenerate.

The pattern matches the simulation and CAP results (Figure S4; full numbers in Table S1). Rogan-Gladen fails by 12–19pp and SLD comparably; Classify & Count and isotonic regression show moderate but growing error (up to 8pp); IPW is accurate on six of eight targets (≤ 0.4 pp in-distribution, ≤ 2.8 pp out-of-distribution) but collapses on the in-distribution young-skewed (-10.1 pp) and out-of-distribution old-skewed (-15.5 pp) targets. Multicalibration achieves ≤ 0.27 pp error across all in-distribution scenarios, including the bimodal shift, and degrades only modestly out of distribution (0.88–1.35pp), reflecting geographic shift along a dimension the calibration set did not span. Across both applications the story is consistent: multicalibration is near-exact when the target’s features lie within the calibration support and degrades predictably when they do not: the scope condition has visible, interpretable bite rather than silent failure.

Why IPW collapses. IPW and MCGrad use the same learner (gradient-boosted trees, default settings, no tuning). For IPW it estimates the propensity that an observation belongs

to the target rather than the calibration set; the weights are the odds $p/(1-p)$, cross-fitted over five folds. The calibration set ($n \approx 644,000$) is about thirty times larger than each target ($n = 20,000$), so the propensities are small on average, and a few calibration observations placed in near-pure target leaves receive odds thousands of times the mean. On the two failing targets the effective sample size of the weights, $(\sum w)^2 / \sum w^2$, falls to 518 (young-skewed) and 2 (out-of-distribution old-skewed), and the estimate is effectively the label of a handful of respondents. IPW can therefore match multicalibration within support, but only with per-target monitoring of the weights and tuning when they degenerate. MCGrad needs neither, because it adjusts predictions bounded in $[0, 1]$ and averages them over every target observation, so no small set of observations can dominate the estimate, and its boosting rounds are chosen by cross-validated early stopping on the calibration data.

Table S1: ACS Employment Prevalence Estimation Error

Setting	Age dist.	True prev.	Raw	CC	RG	PACC	SLD	IPW	Iso.	MCGrad
In-Dist	Original	46.0%	-0.31 [-0.6, +0.3]	-0.07 [-0.5, +0.4]	+0.26 [-0.9, +0.8]	-0.07 [-0.9, +0.6]	+0.01 [-0.9, +0.6]	-0.38 [-0.6, +0.3]	-0.30 [-0.6, +0.3]	-0.27 [-0.6, +0.3]
In-Dist	Young-skewed	12.8%	+1.93 [+1.7, +2.3]	+2.47 [+2.2, +2.9]	-12.82 [-13.5, -12.5]	-12.82 [-13.5, -12.5]	-12.82 [-13.1, -12.2]	-10.12 [-13.3, +8.6]	+2.04 [+1.8, +2.4]	-0.11 [-0.3, +0.2]
In-Dist	Old-skewed	16.8%	+7.23 [+6.8, +7.5]	-6.62 [-7.3, -6.4]	-16.77 [-17.4, -16.4]	-16.77 [-17.4, -16.4]	-16.76 [-17.4, -16.4]	+0.24 [-17.1, +50.0]	+6.65 [+6.2, +7.0]	+0.22 [-0.3, +0.4]
In-Dist	Bimodal	21.1%	+4.57 [+4.2, +4.8]	-1.50 [-1.9, -1.2]	-18.62 [-19.2, -18.1]	-19.97 [-20.5, -19.4]	-16.79 [-17.6, -16.4]	+0.35 [-0.1, +0.5]	+4.33 [+3.9, +4.6]	+0.12 [-0.3, +0.3]
OOD	Original	45.1%	+1.15 [+0.8, +1.8]	+1.40 [+1.1, +2.3]	+2.12 [+1.6, +3.3]	+2.13 [+1.3, +2.9]	+2.25 [+1.4, +3.0]	+1.38 [+0.8, +1.8]	+1.17 [+0.8, +1.8]	+1.35 [+1.0, +1.9]
OOD	Young-skewed	13.0%	+2.93 [+2.5, +3.1]	+3.73 [+3.5, +4.1]	-12.96 [-13.3, -12.7]	-12.96 [-13.3, -12.7]	-12.39 [-12.8, -11.9]	-2.80 [-13.2, +42.7]	+3.08 [+2.7, +3.2]	+0.88 [+0.5, +1.0]
OOD	Old-skewed	16.0%	+8.47 [+7.9, +8.7]	-5.64 [-6.1, -5.1]	-15.97 [-16.6, -15.5]	-15.97 [-16.6, -15.5]	-15.97 [-16.6, -15.5]	-15.45 [-16.2, +75.6]	+7.91 [+7.4, +8.2]	+1.01 [+0.6, +1.4]
OOD	Bimodal	20.8%	+5.91 [+5.5, +6.2]	+0.09 [-0.6, +0.3]	-16.14 [-17.3, -15.9]	-17.27 [-18.1, -16.8]	-15.63 [-16.4, -15.2]	+1.39 [+1.0, +1.6]	+5.69 [+5.4, +6.0]	+1.13 [+0.9, +1.4]

Prevalence estimation error in percentage points (pp) under synthetic age distribution shift. Raw = uncalibrated averaging, CC = Classify & Count, RG = Rogan-Gladen, IPW = importance-weighted prevalence estimation (cross-fitted default LightGBM propensity model on all 16 features), Iso. = Isotonic regression. Brackets: 95% percentile intervals from 50 bootstrap replicates (Section S1.9).

Table S2: CAP Law & Crime Prevalence Estimation Error (Claude Opus 4.6)

Scenario	Shift type	True prev.	CC	RG	SLD	IPW	Iso.	MC (binary)	MC (scores)
Baseline	None	7.9%	+2.2 [+1.6, +2.8]	+0.5 [-0.2, +1.2]	+0.1 [-0.5, +0.8]	+0.2 [-1.1, +1.2]	+0.1 [-0.5, +0.6]	+0.1 [-0.5, +0.7]	+0.2 [-0.4, +0.6]
Country shift	Within-cal.	8.4%	+3.3 [+2.4, +4.0]	+1.7 [+0.6, +2.6]	+1.4 [+0.8, +2.7]	+0.1 [-1.3, +1.5]	+0.9 [+0.3, +1.7]	+0.4 [-0.5, +1.1]	+0.4 [-0.4, +1.0]
Doc-type shift	Within-cal.	6.3%	+1.6 [+1.0, +2.3]	-0.3 [-1.1, +0.5]	-0.6 [-1.2, +0.1]	+0.2 [-1.0, +1.0]	+0.0 [-0.6, +0.6]	-0.0 [-0.7, +0.7]	+0.1 [-0.5, +0.7]
Spain media	OOD doc type	19.5%	+3.6 [+2.7, +4.4]	+3.1 [+2.0, +4.1]	+3.8 [+2.4, +5.3]	-11.9 [-13.0, -10.2]	-2.5 [-3.5, -1.5]	-1.9 [-4.8, -0.5]	-4.5 [-6.6, -2.6]
Belgium TV	OOD doc type	11.1%	+4.8 [+4.3, +5.4]	+3.7 [+2.9, +4.5]	+4.7 [+3.9, +5.4]	-2.4 [-3.5, -1.4]	+2.6 [+1.8, +3.2]	+0.7 [-0.8, +2.0]	+1.6 [+0.6, +2.6]

CC = Classify & Count (fraction of Yes labels); *RG* = Rogan-Gladen adjustment on binary labels; *SLD* = Saerens-Latinne-Decaestecker (label shift, applied to isotonic-recalibrated probability scores); *IPW* = importance-weighted estimation (target-specific density ratio from a cross-fitted gradient-boosted propensity model); *Iso.* = isotonic regression on probability scores; *MC (binary)* = MCGrad on binary labels with base-rate initialization; *MC (scores)* = MCGrad on probability scores. Brackets: 95% percentile intervals from 200 bootstrap replicates (Section S1.9).

S4. Simulation: Design and RMSE

Design. Each document has a binary feature $X \in \{0, 1\}$ and a content signal $U \sim N(0, 1)$ independent of X . The outcome follows $P(Y = 1 | X, U) = \sigma(a_X + 1.5U)$ with $a_0 = -2$ and $a_1 = 1.5$, so $P(Y = 1 | X = 0) \approx 0.19$ and $P(Y = 1 | X = 1) \approx 0.75$; this relationship is the same in every population, and only $P(X)$ shifts. The classifier’s score is $\sigma(a_X + 1.5U + \delta_X)$ with $\delta_0 = 0.8$ and $\delta_1 = 0$: it is calibrated for $X = 1$ documents and overstates the probability for $X = 0$ documents. Each run draws a labeled calibration sample of 10,000 documents at $P(X = 0) = 0.5$, fits every estimator once, and applies it to fresh unlabeled targets of 10,000 documents on a grid of 20 values of $P(X = 0)$ between 0.01 and 0.99. Classify & Count is evaluated at the default 0.5 threshold and at the threshold that reproduces the calibration prevalence; Rogan-Gladen uses 0.5 (Section S1). We report averages over 50 runs (`simulation/helpers.py`).

RMSE. The main-text simulation (Figure 1) reports bias across the shift gradient. Here we add the root mean squared error for Classify & Count at the 0.5 threshold, Rogan-Gladen, isotonic recalibration, and MCGrad.

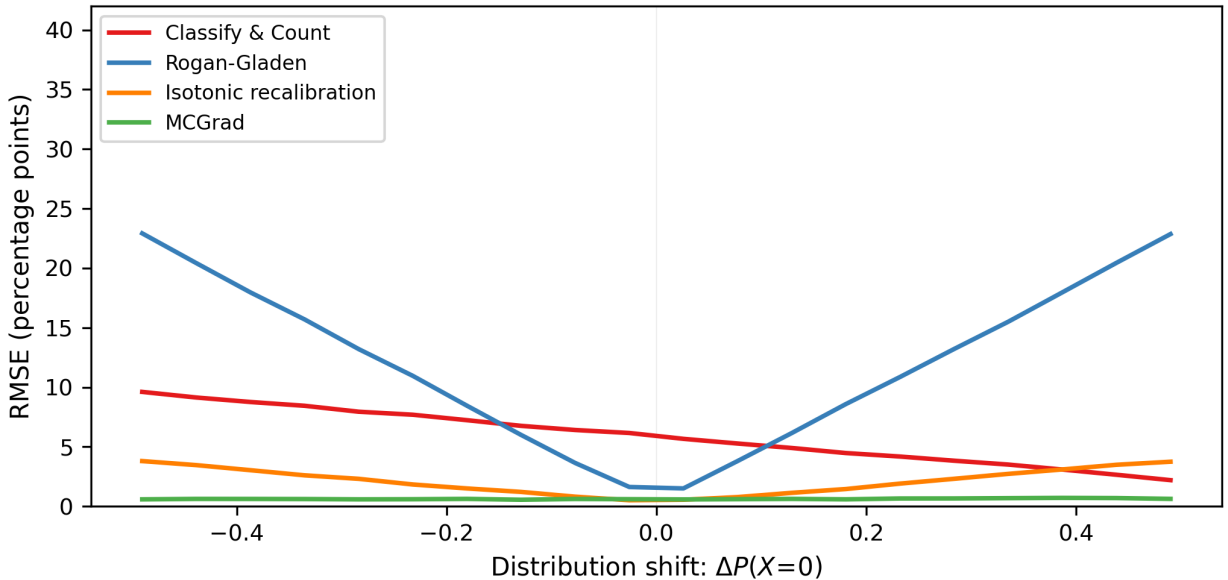


Figure S1: Root mean squared error (RMSE) under covariate shift for Classify & Count at the 0.5 threshold, Rogan-Gladen, isotonic recalibration, and MCGrad, averaged over 50 simulation runs. RMSE closely tracks absolute bias for all methods, confirming that variance

is small relative to bias at this sample size. MCGrad has the lowest RMSE once the target departs from the calibration distribution.

S5. Simulation: All Methods

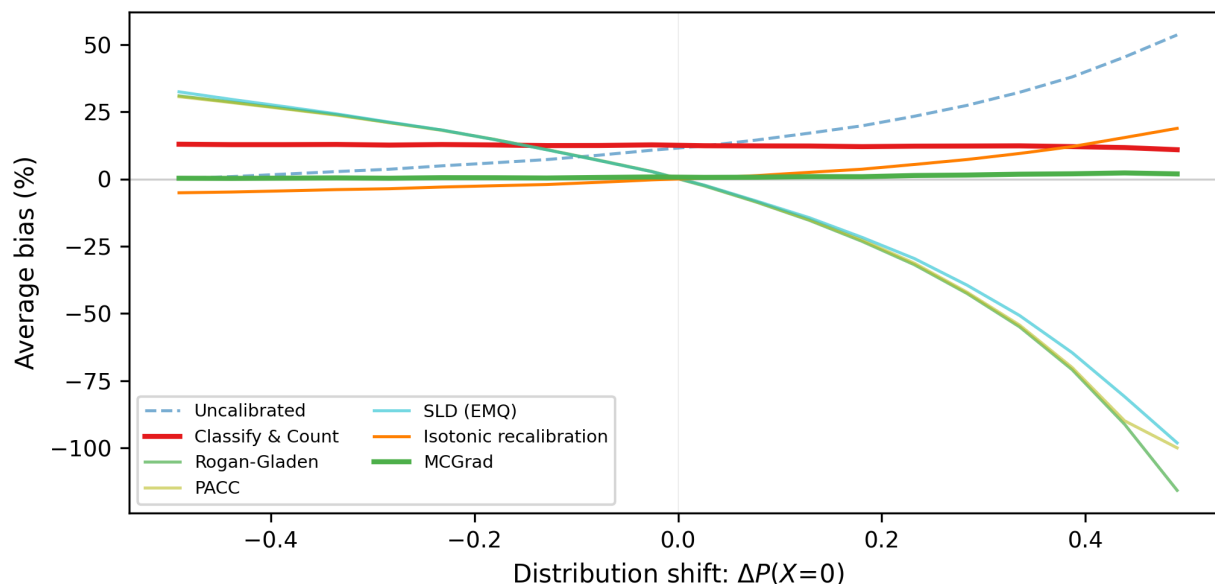


Figure S2: Simulation bias curves for all seven methods (uncalibrated averaging, Classify & Count, Rogan-Gladen, PACC, SLD/EMQ, isotonic recalibration, MCGrad). Rogan-Gladen and PACC fail badly under shift: at the most extreme shift Rogan-Gladen passes -100% (a negative prevalence) and PACC, truncated to $[0, 1]$, sits at the -100% floor. SLD, run on isotonic-recalibrated scores, is unbiased at the calibration distribution but tracks Rogan-Gladen under shift (about $+32\%$ and -98% at the two extremes), because it attributes the change in the score distribution to a change in the class prior. Isotonic recalibration is unbiased at the calibration distribution and drifts to about $+19\%$ at the most extreme shift. MCGrad stays within about 2% throughout.

S6. Claude Opus Score Distribution

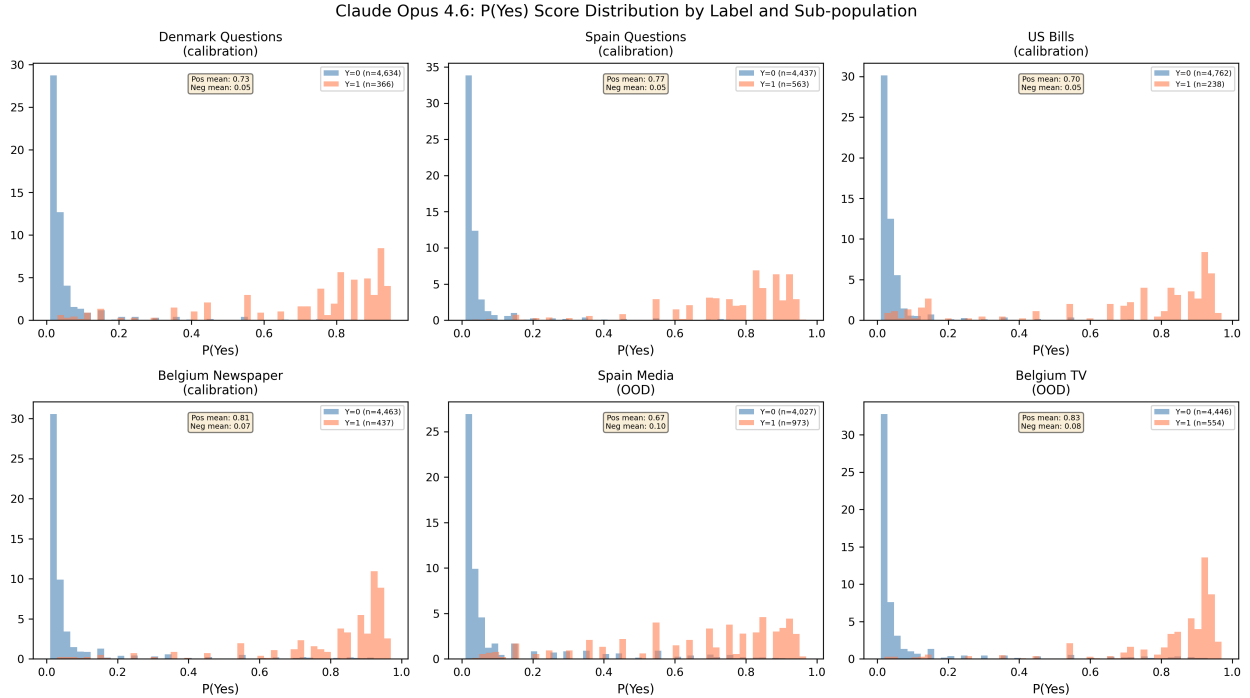


Figure S3: Claude Opus 4.6 $P(\text{Yes})$ score distribution by label across six CAP sub-populations. Scores are well-separated (mean 0.75 for positives vs. 0.07 for negatives) with 43 unique values and no boundary mass.

For reference, the discrimination figures reported in the main text are tabulated here. Pooled over the four in-distribution sub-populations, AUC is 0.959 for the binary (Yes/No) condition and 0.987 for the probability-score condition (0.940 and 0.981 over all six); per-language AUCs of the probability scores range from 0.978 to 0.985. The probability-score condition has 43 unique values (this figure); the verbalized-confidence elicitation used in the Llama replication produces 11 unique values (SI Section S2). All discrimination metrics are computed against the CAP expert codes as the gold standard.

S7. Relationship to the ReadMe Family of Quantifiers

The political-science quantification literature is anchored by ReadMe (Hopkins and King 2010) and its successor ReadMe2 (Jerzak et al. 2023), which estimate category proportions directly from document-feature distributions without per-document classification. Classic ReadMe, like the Saerens-Latinne-Decaestecker (SLD) algorithm (Saerens et al. 2002) we report in Tables S1–S2 and many of the quantifiers surveyed by González et al. (2017), assumes *label shift*: the class-conditional feature distribution $P(X | Y)$ is stable across the labeled and target sets while the class prior $P(Y)$ may change. ReadMe2 relaxes this by choosing feature summaries that are more stable between labeled and target sets; we do not analyze its identifying restriction formally here, and like the others it is estimated for a given labeled/target pair.

The regime studied in the main text is the opposite: *covariate shift*, where $P(X)$ changes across populations while $P(Y | X)$ is stable. Which assumption fits is sometimes argued from causal direction (Schölkopf et al. 2012). If the category causes the features, as a topic arguably causes the words used to discuss it, then $P(X | Y)$ is the stable part and label shift is the natural model. If the features cause the category, $P(Y | X)$ is stable and covariate shift is natural. For measurement, what matters more is how the target differs from the source. When the target is formed by selecting on features (a different country, venue, period, or document type, or a corpus that simply contains more of one kind of document), the chance that a given document belongs to the category does not change, so $P(Y | X)$ is stable whatever the causal direction. The within-category feature distribution does change, because the mix of documents within each category changes: a law-and-crime document from a corpus heavy in Belgian TV reads differently from one in a corpus of US bills. The label-shift identity that ReadMe solves is then misspecified. Label shift is the better model when the target differs only in how common the category is, for example when more law-and-crime events occur but each is written about the same way.

We test this directly by benchmarking ReadMe on the CAP corpora, using the same calibration set and shift scenarios as Table S2. We report two implementations on the identical documents: a from-scratch implementation of the Hopkins-King estimator (binary word-presence features summarized over 300 random feature subsets of size 15; `cap_analysis/readme_baseline.py`), and the official IQSS `readme` package’s classic estimator (`readme0`; 200 subsets, vocabulary 2,500). Both are classifier-free: they use the document text and the calibration labels, not the LLM’s outputs.

Scenario	Shift	True	ReadMe (ours)	ReadMe (official <code>readme0</code>)
Baseline	none	7.9%	+0.8	+5.3
Country shift	within-cal.	8.4%	+67.6	+60.1
Doc-type shift	within-cal.	6.3%	+7.1	-5.0
Spain media	OOD doc type	19.5%	+4.5	+44.5
Belgium TV	OOD doc type	11.1%	+13.5	+51.5

Table S4: ReadMe prevalence-estimation error (pp) on the CAP corpora; identical calibration set and scenarios as Table S2. Both estimators are classifier-free (text only). For comparison, MCGrad (Table S2) stays within 1.9pp across all scenarios.

The two implementations differ in magnitude, and on the document-type shift even in sign, which depends on vocabulary, feature weighting, and profile-matching choices. ReadMe is near-exact *only* without shift (our implementation attains +0.8pp at baseline, validating it, while the official `readme0` is already +5 to +7pp at baseline on this short, multilingual corpus) and is inaccurate and highly unstable under every shift, severely so out of support (the official estimator reaches +44 to +52pp on the held-out genres; the within-calibration country shift exceeds +60pp in both implementations). The component estimates are also unstable: across the 300 random feature subsets, the standard deviation of our implementation’s subset-level estimates is 11pp at baseline and 32–42pp under shift. This is the dispersion

of the components, not the uncertainty of their average. The mechanism is interpretable: ReadMe’s binary word-presence profiles are sensitive to document length, so when the target’s mean feature density departs from the calibration set’s (long Belgian newspaper documents activate roughly twice as many features as the calibration average; short Spanish-media and Belgian-TV snippets far fewer), the within-class feature distribution shifts and the label-shift identity breaks. This is the covariate-shift failure the theory predicts, shown directly for the best-known political-science quantifier rather than inferred from SLD.

ReadMe2 (Jerzak et al. 2023) refines the feature summary (word-vector “vector summaries” and a learned matched projection) but still infers proportions from category-conditional feature summaries estimated on the labeled set; whether its identifying restriction holds under the covariate shift studied here has not been established. We installed and ran the official ReadMe2 estimator; a faithful evaluation requires its **undergrad** vector-summary features, and when supplied instead with multilingual-BERT document embeddings the estimator erred by roughly +20pp even at the no-shift baseline, mis-specified for that feature representation and the rare (~8%) positive class, so we do not report it as a fair head-to-head comparison. The defensible empirical statement is the classic-ReadMe result in Table S4.

S8. Calibration Diagnostics

Table S5 reports calibration before and after MCGrad for every analysis, on held-out data only: the in-distribution test split and, for ACS and CAP, the out-of-distribution sets. ECCE is the estimated cumulative calibration error of Arrieta-Ibarra et al. (2022), the range of the running sum of label-minus-score differences over observations sorted by score; we report it in percentage points and in units of its standard deviation under perfect calibration (σ). MCE (Guy et al. 2025) is the largest ECCE σ over subgroups formed from the features MCGrad uses, up to three-way intersections (default `mcgrad.metrics` settings); its absolute value rescales that maximum by the global standard deviation. The pre-calibration score is the classifier or LLM score, except for CAP Opus binary, where MCGrad’s input is the constant calibration base rate and the LLM label enters as a feature. For that variant the pre-calibration global ECCE only reflects arbitrary ordering among tied scores (and, out of distribution, the shift in prevalence), while its MCE reflects the ignored label.

Within the calibration distribution, MCGrad brings global ECCE to $1.0\text{--}2.0\sigma$ in every analysis, consistent with perfect calibration, and the worst subgroup to at most 0.71pp. The ACS in-distribution MCE of 6.2σ corresponds to 0.22pp and is detectable only because the test set has about 920,000 observations. Out of distribution, calibration degrades in line with the prevalence error in Tables S1–S3. Because the running sum ends at the mean label-minus-score difference, ECCE bounds the absolute error of the averaged prediction, and the post-calibration ECCE on the out-of-distribution sets (0.75–5.09pp) is close to the corresponding MCGrad error. For Opus scores on Spain media and for the ACS held-out states, MCGrad makes global calibration worse (1.74 to 4.57pp and 1.25 to 1.46pp).

Analysis	Evaluation set	ECCE (pp)	ECCE σ	MCE (pp)	MCE σ
Simulation	In-dist. test	5.78 $\rightarrow$ 0.71	15.1 $\rightarrow$ 2.0	7.67 $\rightarrow$ 0.71	20.1 $\rightarrow$ 2.0
ACS	In-dist. test	0.85 $\rightarrow$ 0.06	22.0 $\rightarrow$ 1.7	3.46 $\rightarrow$ 0.22	89.2 $\rightarrow$ 6.2
ACS	OOD states	1.25 $\rightarrow$ 1.46	57.0 $\rightarrow$ 71.8	3.75 $\rightarrow$ 1.47	170.8 $\rightarrow$ 72.2
Opus, binary	In-dist. test	0.93 $\rightarrow$ 0.19	2.8 $\rightarrow$ 1.0	21.34 $\rightarrow$ 0.67	63.5 $\rightarrow$ 3.6
Opus, binary	Spain media	11.41 $\rightarrow$ 2.48	29.6 $\rightarrow$ 8.0	30.84 $\rightarrow$ 8.34	80.1 $\rightarrow$ 27.0
Opus, binary	Belgium TV	3.08 $\rightarrow$ 0.75	8.0 $\rightarrow$ 2.8	23.68 $\rightarrow$ 0.93	61.5 $\rightarrow$ 3.4
Opus, scores	In-dist. test	3.34 $\rightarrow$ 0.30	12.4 $\rightarrow$ 1.7	3.34 $\rightarrow$ 0.57	12.4 $\rightarrow$ 3.3
Opus, scores	Spain media	1.74 $\rightarrow$ 4.57	4.4 $\rightarrow$ 14.3	1.91 $\rightarrow$ 4.57	4.9 $\rightarrow$ 14.3
Opus, scores	Belgium TV	5.45 $\rightarrow$ 1.78	17.2 $\rightarrow$ 7.3	5.45 $\rightarrow$ 1.78	17.2 $\rightarrow$ 7.3
Llama, scores	In-dist. test	18.10 $\rightarrow$ 0.22	88.0 $\rightarrow$ 1.4	18.10 $\rightarrow$ 0.46	88.0 $\rightarrow$ 3.0
Llama, scores	Spain media	16.17 $\rightarrow$ 5.09	93.7 $\rightarrow$ 32.2	16.17 $\rightarrow$ 5.09	93.7 $\rightarrow$ 32.2
Llama, scores	Belgium TV	14.04 $\rightarrow$ 3.44	68.8 $\rightarrow$ 18.4	14.04 $\rightarrow$ 3.44	68.8 $\rightarrow$ 18.4

Table S5: Calibration before $\rightarrow$ after MCGrad on held-out data; Opus and Llama rows are the CAP analyses. ECCE = estimated cumulative calibration error (Arrieta-Ibarra et al. 2022); MCE = multicalibration error, the maximum ECCE over feature-defined subgroups (Guy et al. 2025); pp = percentage points; σ = standard deviations under perfect calibration. Simulation: one calibration and one evaluation sample of 10,000 at $P(X = 0) = 0.5$, subgroups on X . ACS subgroups on all 16 features; CAP subgroups on country, document type, party, decade, and (Opus) text length, plus the LLM label for the binary variant. Produced by the run scripts (`paper/images/*_calibration.json`).

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